## MATH 415 - FALL 2020 - EXTRA PRACTICE PROBLEMS - WEEK 8

**1.** *Compute the following determinants using* ***row reduction to echelon form****:*

(1) $\begin{vmatrix} 1 & 5 & -4 \\ -1 & -4 & 5 \\ -2 & -8 & 7 \end{vmatrix}$

(2) $\begin{vmatrix} 1 & 3 & 0 & 2 \\ -2 & -5 & 7 & 4 \\ 3 & 5 & 2 & 1 \\ 1 & -1 & 2 & -3 \end{vmatrix}$

**2.** *Find the determinant of $A$ if*

$$A = S \begin{bmatrix} \lambda_1 & 2 \\ 0 & \lambda_2 \end{bmatrix} S^{-1}.$$

**3.** *Let $A$ be an $n \times n$-matrix with eigenvalue $\lambda$. Which of the following statements are true? Give reasons why!*

(a) $\lambda^2$ is an eigenvalue of $A^2$.
(b) $\lambda^{-1}$ is an eigenvalue of $A^{-1}$.
(c) $\lambda + 1$ is an eigenvalue of $A + I$.

**4.** *True/False. Give reasons why! Below $A$ and $B$ are matrices of an appropriate size.*

(1) Suppose that $A$ and $B$ are both $n \times n$ matrices. Then $\det(A+B) = \det(A) + \det(B)$.
(2) Suppose that $A$ is a $3 \times 3$ matrix with $\det(A) = 9$. Then $\det(2A) = 18$.
(3) Suppose that $A$ is a $2 \times 3$ matrix. Then $\det(A^T A) = 0$.
(4) Suppose that $A$ is a $2 \times 3$ matrix. Then $\det(AA^T) = 0$.
(5) The determinant of $A$ is the product of the pivots in any echelon form $U$ of $A$, multiplied by $(-1)^r$, where $r$ is the number of row interchanges made during row reduction from $A$ to $U$.
(6) A square matrix $A$ is not invertible iff 0 is an eigenvalue of $A$.
(7) The eigenvalues of a square matrix are the numbers on its main diagonal.
(8) If $\mathbf{v}_1$ and $\mathbf{v}_2$ are linearly independent eigenvectors of a square matrix $A$, then they correspond to different eigenvalues.
(9) The sum of two eigenvectors of a matrix $A$ is also an eigenvector of $A$.

**5.** *Let $A$ be a $2 \times 2$-matrix , and suppose that* $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ *is an eigenvector of $A$ with eigenvalue* 0 *and* $\begin{bmatrix} 2 \\ -1 \end{bmatrix}$ *is an eigenvector of $A$ with eigenvalue 1. Determine $A$.*

**6.** *Let $A$ be an $n \times n$-matrix and let $\lambda_1, \lambda_2$ be real numbers such that $\lambda_1 \neq \lambda_2$. Let $\mathbf{x}, \mathbf{y}$ be such that $A\mathbf{x} = \lambda_1 \mathbf{x}$ and $A^T \mathbf{y} = \lambda_2 \mathbf{y}$. Check that $\mathbf{x}$ and $\mathbf{y}$ are orthogonal to each other.*

**7.** *This exercise explores a particular type of matrices, called* ***nilpotent matrices****, which have interesting properties and they are an useful source of counterexamples in mathematics! A square matrix $A$ is nilpotent if there exists a positive integer $k$ such that $A^k = 0$ (i.e. the $k$-th power of the matrix $A$ is the zero matrix).*

(a) Give two examples of non zero nilpotent $3 \times 3$ matrices.

(b) Explain why a nilpotent matrix is non invertible. (Hint: Use determinants).

(c) Explain why all the eigenvalues of a nilpotent matrix $A$ are 0.

**8.** *Let $A, B$ be two $n \times n$-matrices such that $AB = BA$.*

(a) Suppose $\boldsymbol{v}$ is an eigenvector of $A$ with eigenvalue $\lambda$. Is $B\boldsymbol{v}$ an eigenvector of $A$? If so, what is the eigenvalue of that eigenvector?

(b) Suppose $A$ has eigenvectors $\boldsymbol{v_1}, \ldots, \boldsymbol{v_n}$ with distinct eigenvalues $\lambda_1 \neq ... \neq \lambda_n$. Is each $\boldsymbol{v_i}$ also an eigenvector of $B$? (This question is a bit tricker. Hint: Note that each of the eigenspaces of $A$ has dimension 1 and then use your answer to (a).).